

Hylton Reber

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OBJECTIVE

Mechanical engineering graduate seeking an entry-level role to apply design, analysis, and problem-solving skills to support impactful engineering projects and deliver high-quality results within a collaborative team.

PRACTICAL EXPERIENCE

Axes & Axles LLC | Co-Founder & CEO | Red Bull Soapbox Race Denver 2026 *April 2026 – June 2026*

- Re-engineered a donor go-kart chassis by analyzing structural members for load-bearing necessity, shortened the frame 2 ft and removed the roll cage and non-essential bars — reducing weight while maintaining structural integrity.
- Iterated front suspension through two full redesigns to resolve clearance constraints; diagnosed and corrected a steering inversion by extending the steering arm geometry behind the spindle.
- Engineered a field replacement brake system after hydraulic failure 24 hours before competition — solution passed safety inspection and the team competed.

Hyster-Yale Forklift Overhead Guard | Capstone Senior Team Project *August 2024 – May 2025*

- Reduced resonance noise in an overhead guard to contribute to R&D and improve the user experience.
- Integrated accelerometers via Arduino to perform modal testing.
- Interpreted accelerometer sensor specification sheets and configured 8-bit registers in C++.
- Developed a physics-based testing method to determine the natural frequency for modal analysis.

Refrigeration Cycle Team Project | Senior Laboratory *August 2024 – December 2024*

- Conducted experimental analysis on a refrigeration system, collecting 20+ measurement cycles of temperature, pressure, and airflow across condenser and evaporator.
- Applied thermodynamic principles and state postulate analysis with EES and MATLAB to determine system heat transfer and efficiency.

SolidWorks Steam Engine Full Assembly | SolidWorks Course *Spring 2025*

- Modeled and assembled a full steam engine in SolidWorks, producing detailed part drawings, renderings, and assembly animations while applying design documentation, tolerancing, and configuration control per engineering drawing standards.

Machine Shop Experience | UI Machine Shop *Fall 2023 – Spring 2025*

- Accumulated 100+ hours of hands-on machining on manual and CNC mills and lathes.
- Fabricated components (nested cube, tap holder) demonstrating proficiency in measurement, tolerancing, and multi-axis setups.

Student Manager | University of Idaho Outdoor Program – Climbing Center *January 2023 – May 2025*

- Wrote & edited dozens of pages of operational plans and provided ongoing support & development for staff.

EDUCATION: UNIVERSITY OF IDAHO, MOSCOW, ID

Bachelor of Science | Mechanical Engineering *May 2025*

GPA: 3.63

SKILLS & CERTIFICATIONS

Certifications: FE Exam – Passed; Certified SolidWorks Expert (CSWE); Aerospace Academic Certificate.

Proficient: Microsoft Office, MATLAB, Python, EES, CAD, Technical Drawings, and Technical Writing.

Technical Electives: Fatigue and Fracture Mechanics, Gas Dynamics, Turbomachinery, Aerodynamics.

CLUBS & ORGANIZATIONS

Risk Manager | Phi Kappa Tau *August 2022 – May 2025*

Aeromechanical Member | Vandal Atmospheric Science Team (VAST) *September 2021 – May 2024*

Eagle Scout | Scout Troop 310 – Idaho Falls, ID *2013 – 2020*